

- $10^{10} = 10^{10} < \log_2(\log_2 n) < \log_2 \sqrt{n} = \log_2 n = \log_3 n < \log_2 n^{\log_2 n} = (\log_2 n)^2 < n \log_2 n < \sqrt{n} < n^2 < 2^{\log_2 n} = \sqrt{2}^{\log_2 n} < 2 = 32n!$
 10^{10} and 10^{-10} are constant so both have no growth.
 $\log_2(\log_2 n)$ is growing at the smallest rate
 $\log_2 \sqrt{n}$, $\log_2 n$, and $\log_3 n$ are all growing at a logarithmic rate just with different constants and log bases
 $\log_2 n^{\log_2 n} = \log_2 n \cdot \log_2 n = (\log_2 n)^2$ which is less than $n \log_2 n$
 $2^{\log_2 n}$ and $\sqrt{2}^{\log_2 n}$ are both growing at a rate of $\log_2 n$ similarly 2^n and 3^n are growing at a rate of power n and $n!$ is growing at the highest rate

2 a) $\log_2(n!) = \Omega(1)$
 $\log_2 n! \geq C \cdot 1 \quad n > k$
 $\lim_{n \rightarrow \infty} \frac{\log_2 n!}{1} \rightarrow \infty \quad C = \infty \therefore \log_2 n! = \Omega(1)$
 $\log(n!) = O(n \log n)$
 $\lim_{n \rightarrow \infty} \frac{\log n!}{n \log_2 n} \rightarrow \frac{\log \infty}{\infty \log \infty} \rightarrow 0$
 $C = 0 \therefore \log_2 n! = O(n \log n)$

b) $3n^{\sqrt{n}} = \Theta(2^{\sqrt{n} \log_2 n})$
 $\lim_{n \rightarrow \infty} \frac{3n^{\sqrt{n}}}{2^{\sqrt{n} \log_2 n}}$
 $\hookrightarrow \log_2 3n^{\sqrt{n}} = \log_2 3 + \log_2 n^{\sqrt{n}} = \log_2 3 + \sqrt{n} \log_2 n$
 $\log_2 2^{\sqrt{n} \log_2 n} = \sqrt{n} \log_2 n - \log_2 2$
 $\lim_{n \rightarrow \infty} \frac{\log_2 3 + \sqrt{n} \log_2 n}{\sqrt{n} \log_2 n - 1} \rightarrow \frac{\sqrt{n} \log_2 n}{\sqrt{n} \log_2 n} = 1$
 $\therefore C = 1$ meaning $3n^{\sqrt{n}} = \Theta(2^{\sqrt{n} \log_2 n})$

3 $T(n) = 3T(\frac{n}{2}) + n$ $T(n) = 8T(\frac{n}{2}) + n^2$
 $a=3 \quad \log_2 a$ $a=8 \quad \log_2 8 = 3 > 2$
 $b=2 \quad \log_2 3 > 1$ $b=2 \quad \log_2 8 = 3 > 2$
 $f(n)=n \quad \Theta(n^{\log_2 3})$ $f(n)=n^2 \quad \Theta(n^{\log_2 8})$
 $k=1 \quad p=0$ $k=2 \quad p=0 \quad = \Theta(n^3)$

4. $\frac{2^n}{n!} \rightarrow \frac{f(n+1)}{f(n)} = \frac{2^{n+1}}{(n+1)!} = \frac{2 \cdot 2^n}{(n+1) \cdot n!} = \frac{2}{n+1} \cdot \frac{2^n}{n!}$
 $= \frac{2}{n+1}$
 $\lim_{n \rightarrow \infty} \frac{2}{n+1} \rightarrow 0$ Since $n!$ grows at a much larger rate, the result is zero.

6. 2 for loops $\therefore O(n^2)$
 $\text{calc}(n) = \text{calc}(n-1) + \text{calc}(n-2) + \frac{n(n-1)}{2}$
 driving term = $\frac{n(n-1)}{2}$

homogeneous term = $\text{calc}(n-1) + \text{calc}(n-2)$

This is a fibonacci function which

would make the recurrence

equal to the golden ratio

$\text{calc}(n) \Rightarrow r^2 = r + 1 \Rightarrow r^2 - r - 1 = 0$

$x = \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2}$

$x = \frac{1 \pm \sqrt{5}}{2}$

$f(n) = \Theta(x^n) = \Theta\left(\left(\frac{1 + \sqrt{5}}{2}\right)^n\right)$

7. $f(n) = f(n-1) + n$

to find max find n th sum

$\text{max} = \sum_{k=0}^n f(0) + k \quad f(1) = f(0) + 1 \Rightarrow f(0) = 1$

$= \frac{n(n-1)}{2} + 1$

$= \frac{n^2 - n}{2} + 1$

$= \frac{n^2 - n + 2}{2}$

8 pseudocode

for $i=0, \text{length}(A)-k, i++$

currentsum = 0

for $j=0, k, j++$

currentsum = currentsum + A[i+j]

if currentsum > maxsum

maxsum = currentsum

maxavg = maxsum / k

The asymptotic notation of this program is $O(kn)$ or $O(n)$. This program has 2 loops, one goes from 0 to length-k and the other is going from 0 to k, this does not go through the whole array more than once, resulting in the total time complexity being $O(kn)$. Since k is constant this reduces to $O(n)$.